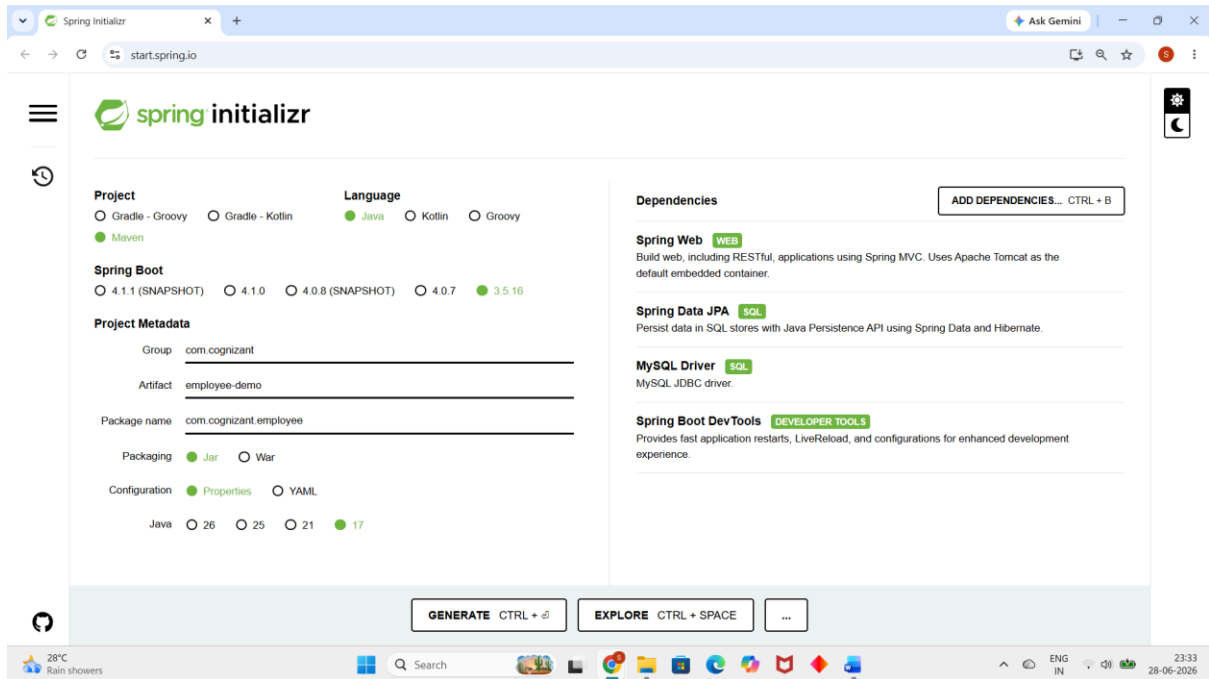


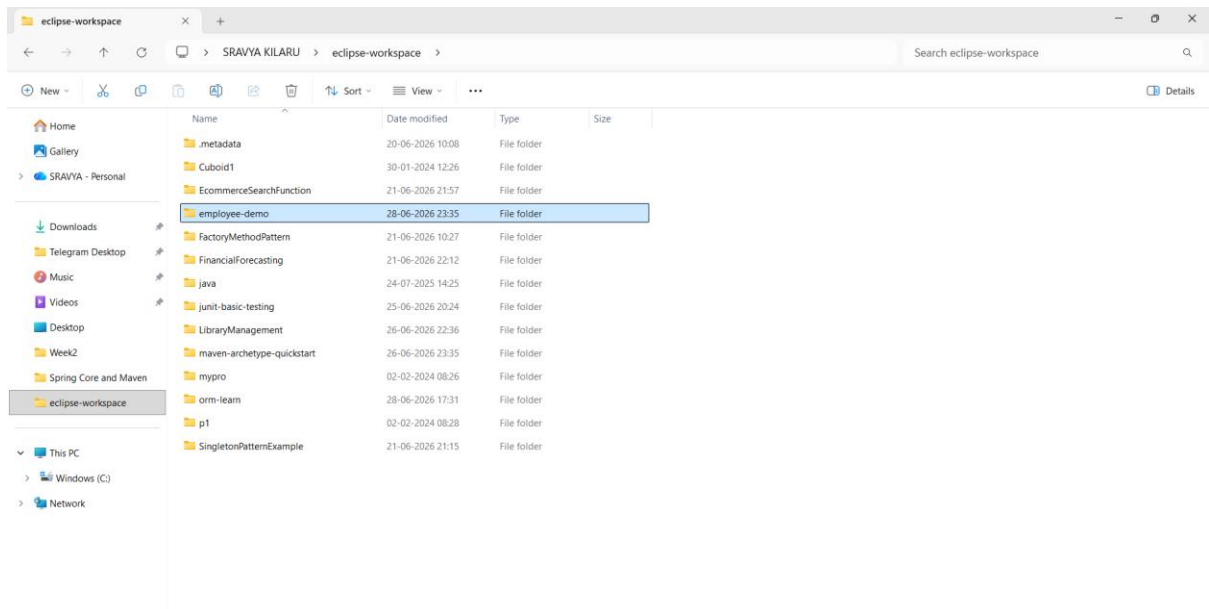
Spring Data JPA with Spring Boot, Hibernate

Difference between JPA, Hibernate and Spring Data JPA

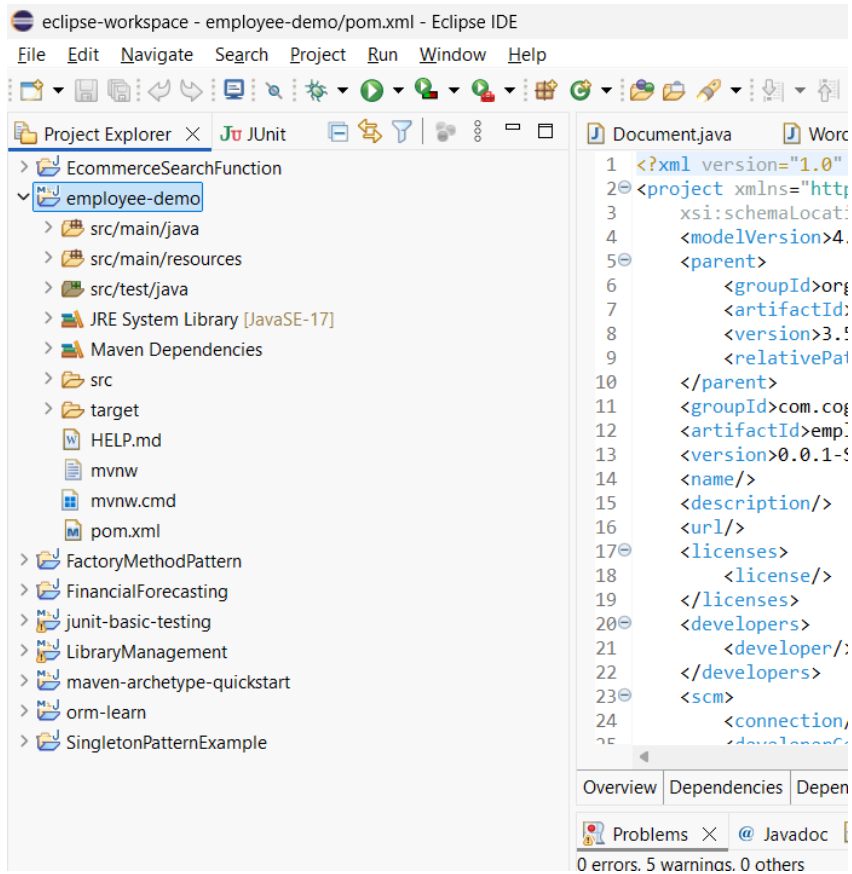
Step 1: Create Spring Boot Project using Spring Initializr



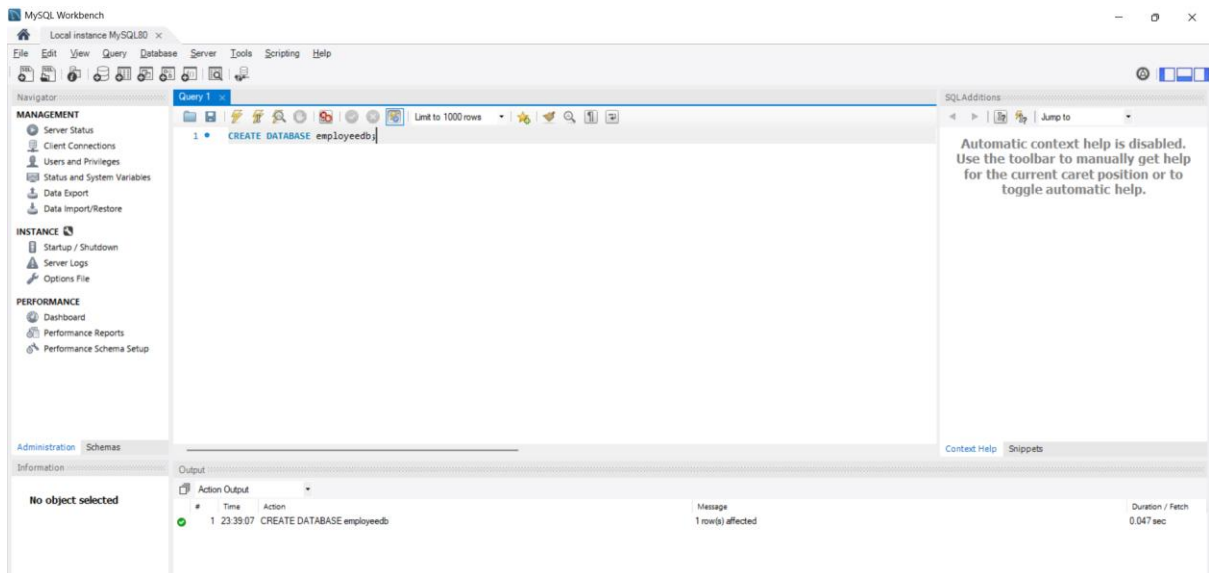
Step 2: Import the Project into Eclipse



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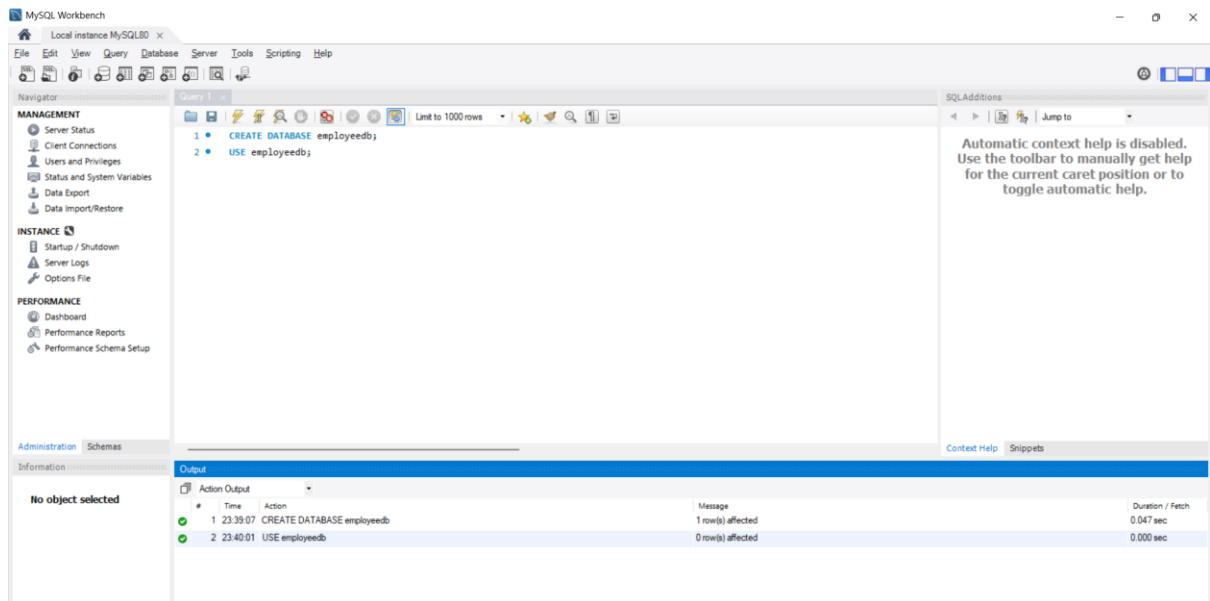


Step 3: Create the MySQL Database

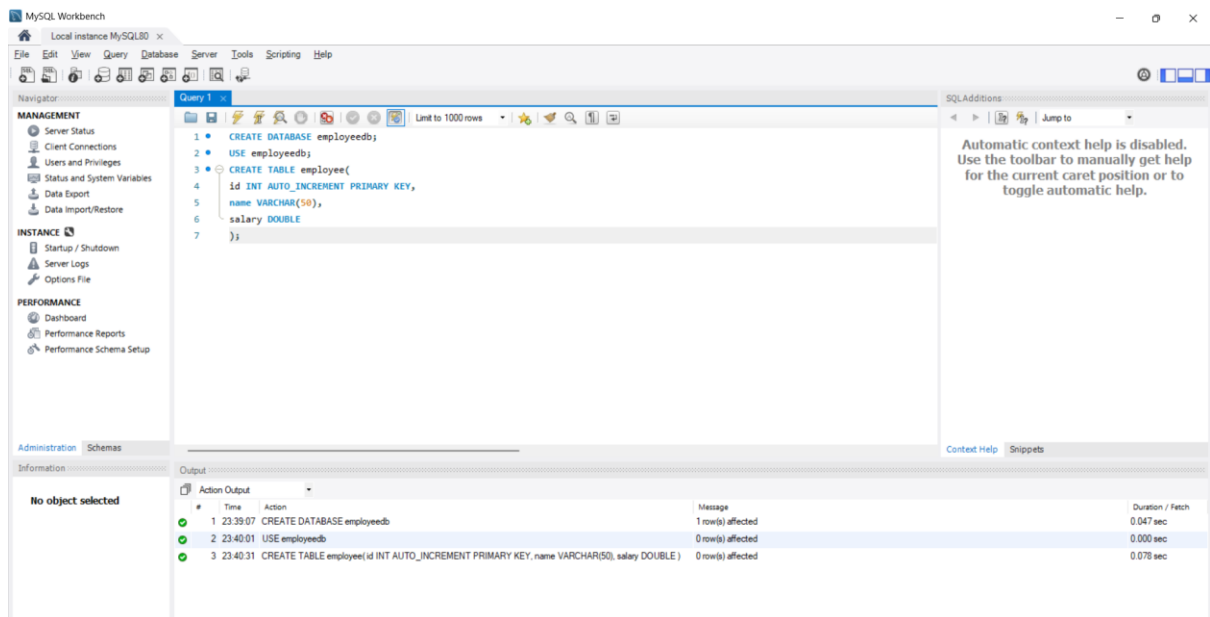


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Step 4: Select the Database



Step 5: Create Employee Table



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Step 6: Verify the Employee Table

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
- Client Connections
- Users and Privileges
- Status and System Variables
- Data Export
- Data Import/Restore

INSTANCE

- Startup / Shutdown
- Server Logs
- Options File

PERFORMANCE

- Dashboard
- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1

```
1 CREATE DATABASE employee;
2 USE employee;
3 CREATE TABLE employee(
4   id INT AUTO_INCREMENT PRIMARY KEY,
5   name VARCHAR(50),
6   salary DOUBLE
7 );
8 DESC employee;
```

Result Grid

Field	Type	Null	Key	Default	Extra
id	int	NO	PRI		auto_increment
name	varchar(50)	YES			
salary	double	YES			

Result 1

Output

Action Output

#	Time	Action	Message	Duration / Fetch
1	23:39:07	CREATE DATABASE employee;	1 row(s) affected	0.047 sec
2	23:40:01	USE employee;	0 row(s) affected	0.000 sec
3	23:40:31	CREATE TABLE employee(id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50), salary DOUBLE)	0 row(s) affected	0.078 sec
4	23:41:38	CREATE DATABASE employee;	Error Code: 1007. Can't create database 'employee'; database exists	0.031 sec
5	23:41:44	DESC employee;	3 row(s) returned	0.015 sec / 0.000 sec

Step 7: Verify Available Tables

MySQL Workbench

Local instance MySQL80

File Edit View Query Database Server Tools Scripting Help

Navigator

MANAGEMENT

- Server Status
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- Performance Reports
- Performance Schema Setup

Administration Schemas

Information

No object selected

Query 1

```
1 CREATE DATABASE employee;
2 USE employee;
3 CREATE TABLE employee(
4   id INT AUTO_INCREMENT PRIMARY KEY,
5   name VARCHAR(50),
6   salary DOUBLE
7 );
8 DESC employee;
9 SHOW TABLES;
```

Result Grid

Tables_in_employee
employee

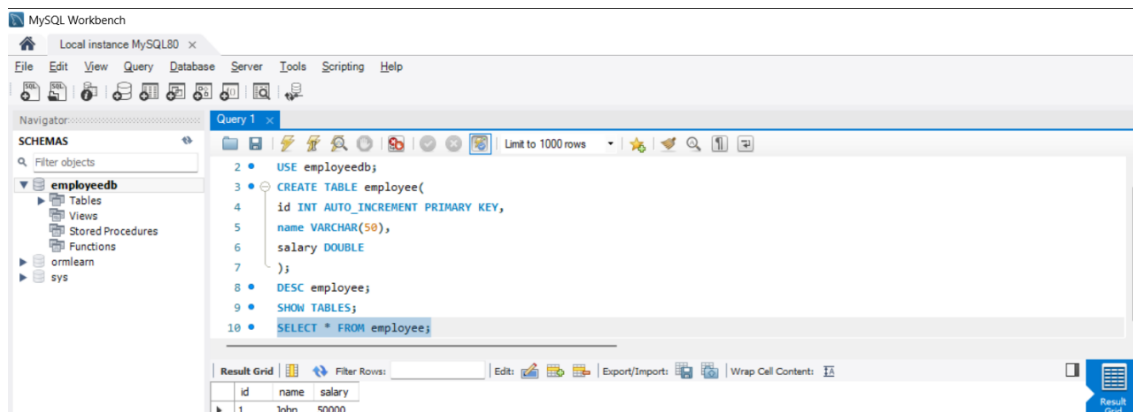
Result 2

Output

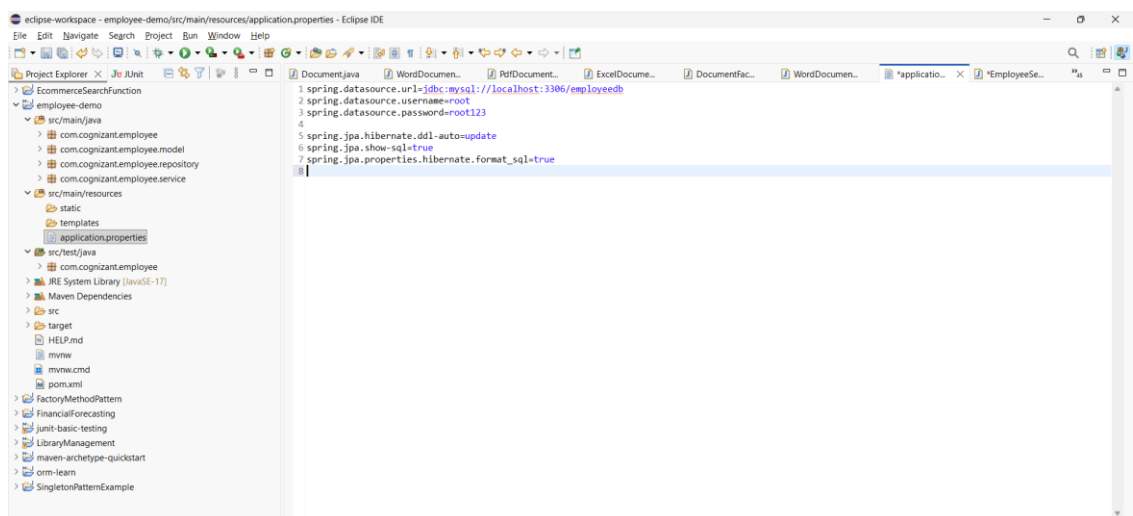
Action Output

#	Time	Action	Message	Duration / Fetch
1	23:39:07	CREATE DATABASE employee;	1 row(s) affected	0.047 sec
2	23:40:01	USE employee;	0 row(s) affected	0.000 sec
3	23:40:31	CREATE TABLE employee(id INT AUTO_INCREMENT PRIMARY KEY, name VARCHAR(50), salary DOUBLE)	0 row(s) affected	0.078 sec
4	23:41:38	CREATE DATABASE employee;	Error Code: 1007. Can't create database 'employee'; database exists	0.031 sec
5	23:41:44	DESC employee;	3 row(s) returned	0.015 sec / 0.000 sec
6	23:42:22	SHOW TABLES;	1 row(s) returned	0.000 sec / 0.000 sec

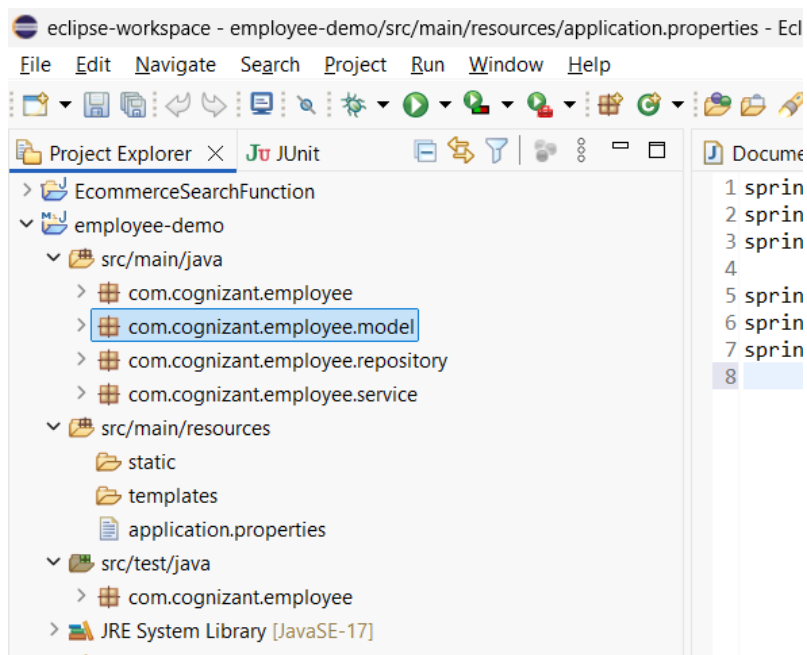
Step 8: Verify Employee Records



Step 9: Configure application.properties

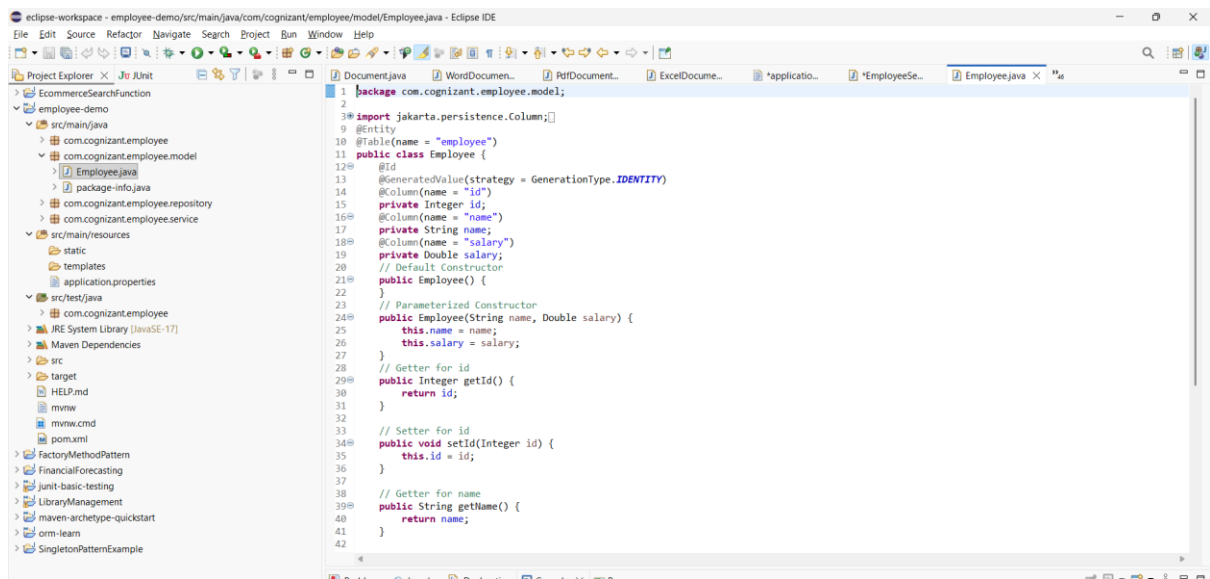


Step 10: Create the Model Package

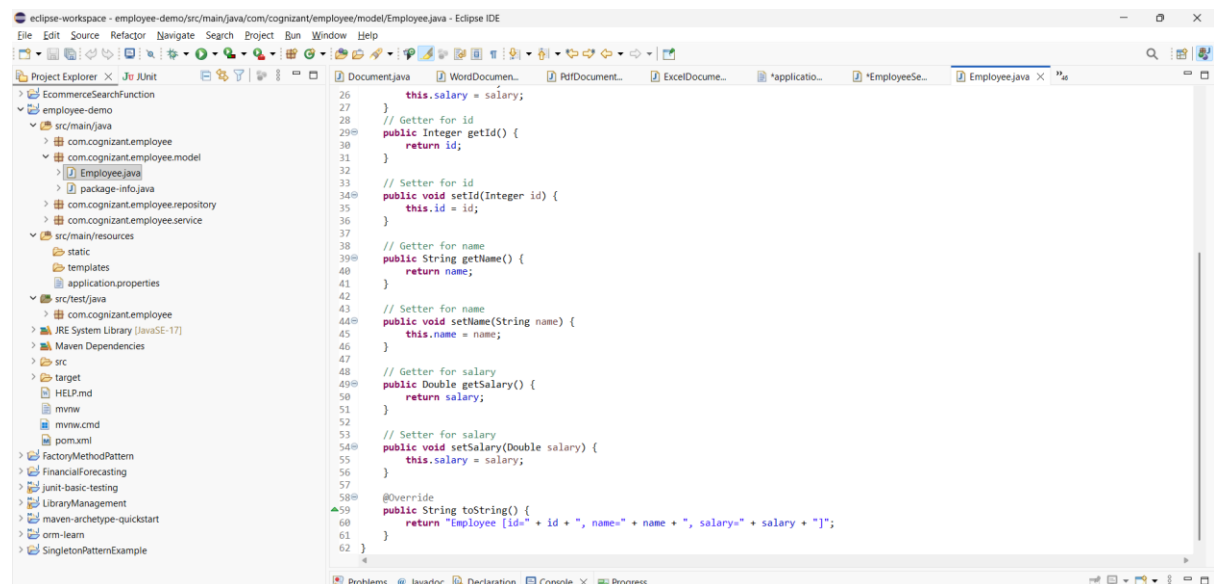


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Step 11: Create Employee Entity Class

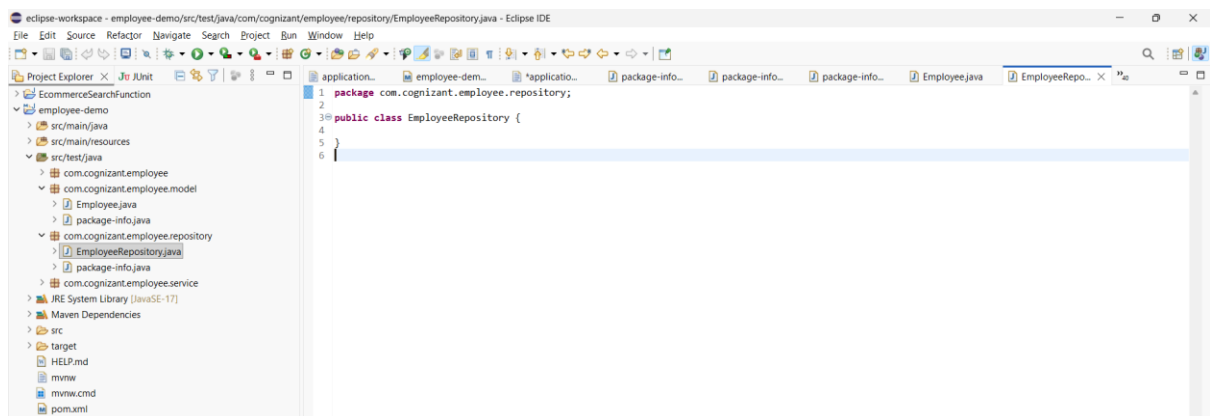


```
1 package com.cognizant.employee.model;
2
3 import jakarta.persistence.Column;
4
5 @Entity
6 @Table(name = "employee")
7 public class Employee {
8     @Id
9     @GeneratedValue(strategy = GenerationType.IDENTITY)
10    @Column(name = "id")
11    private Integer id;
12    @Column(name = "name")
13    private String name;
14    @Column(name = "salary")
15    private Double salary;
16    // Default Constructor
17    public Employee() {
18    }
19    // Parameterized Constructor
20    public Employee(String name, Double salary) {
21        this.name = name;
22        this.salary = salary;
23    }
24    // Getter for id
25    public Integer getId() {
26        return id;
27    }
28    // Setter for id
29    public void setId(Integer id) {
30        this.id = id;
31    }
32    // Getter for name
33    public String getName() {
34        return name;
35    }
36    // Setter for name
37    public void setName(String name) {
38        this.name = name;
39    }
40    // Getter for salary
41    public Double getSalary() {
42        return salary;
43    }
44    // Setter for salary
45    public void setSalary(Double salary) {
46        this.salary = salary;
47    }
48    @Override
49    public String toString() {
50        return "Employee [id=" + id + ", name=" + name + ", salary=" + salary + "];";
51    }
52 }
```



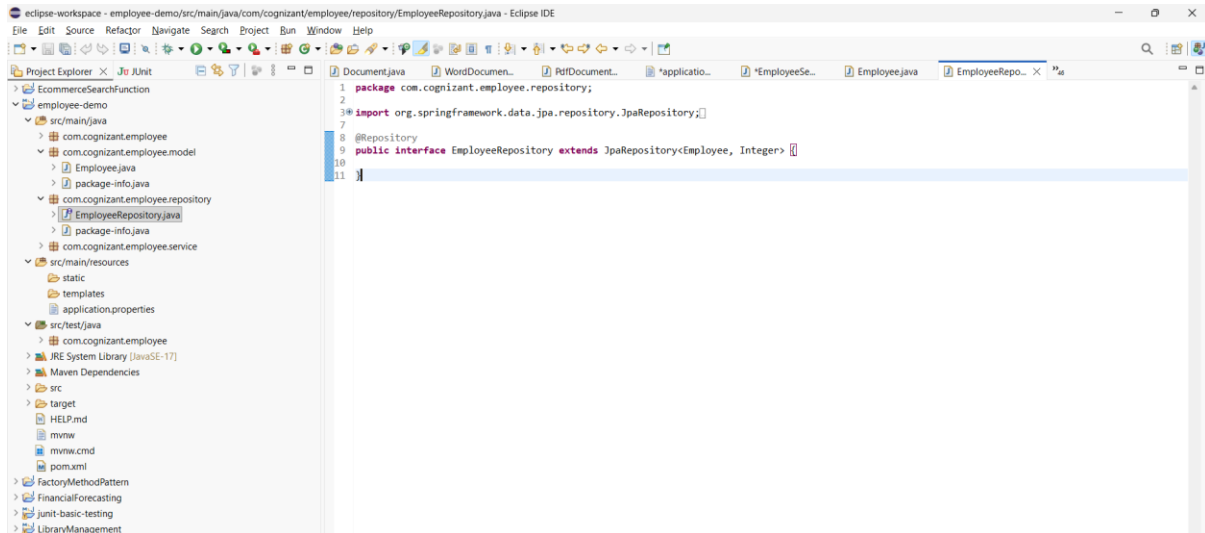
```
1 package com.cognizant.employee.repository;
2
3 public class EmployeeRepository {
4
5 }
6 }
```

Step 12: Create Repository Interface

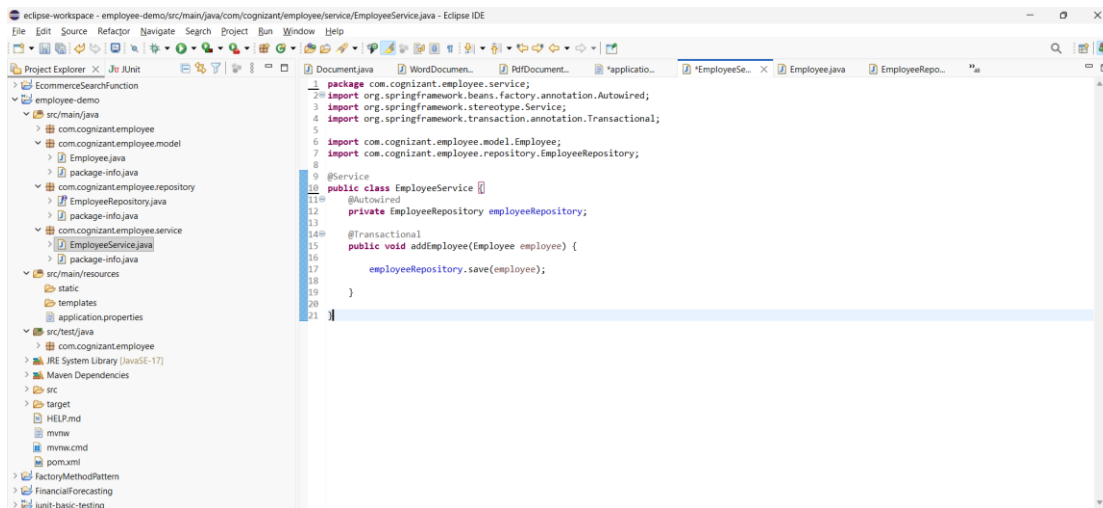


```
1 package com.cognizant.employee.repository;
2
3 public class EmployeeRepository {
4
5 }
6 }
```

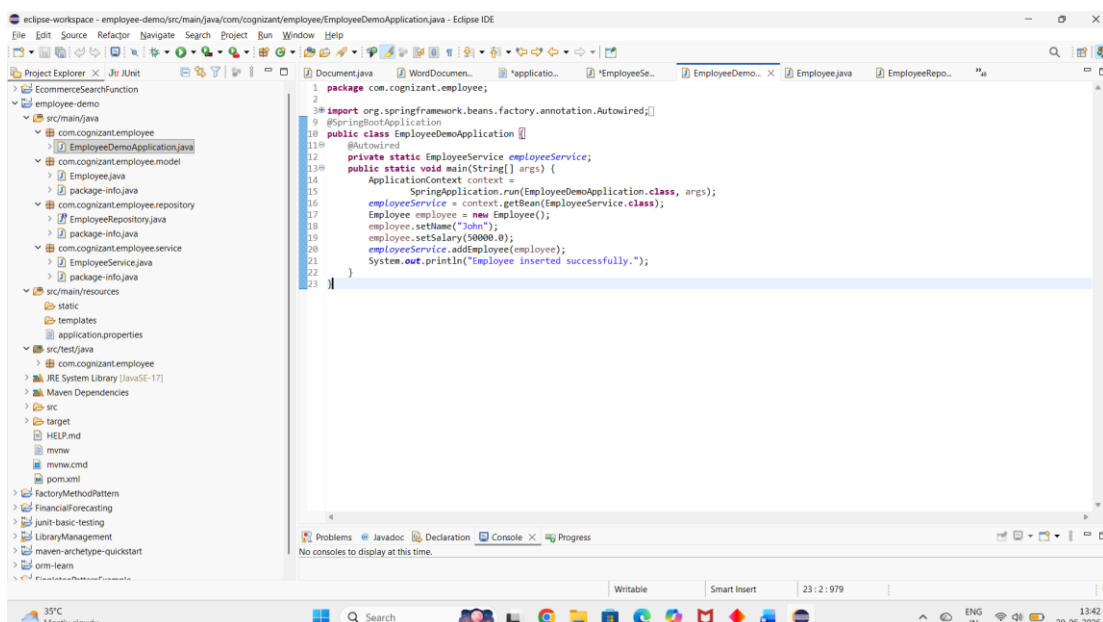
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Step 13: Create Service Class



Step 14: Modify Main Application



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Step 15: Execute the Application and Verify the Console Output

